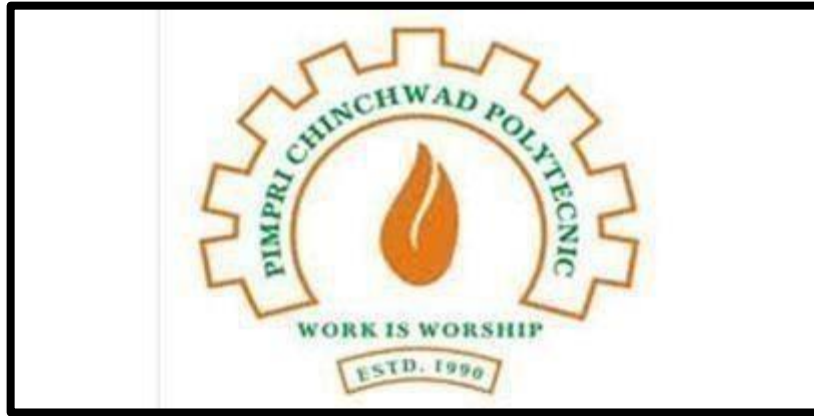


A REPORT ON

“INPLANT TRAINING”



DIPLOMA IN COMPUTER ENGINEERING

AFFILIATED TO MSBTE

Submitted By

Atharva Bakwad

Under the guidance

Mrs. S.G.Aparanji

**DEPARTMENT OF COMPUTER ENGINEERING
PIMPRI CHINCHWAD POLYTECHNIC AKURDI, PUNE.**

2026-27

Certificate of Completion of Industrial Training



Maharashtra State Board of Technical Education

Certificate of Completion

of Industrial Training

This is to certify that

Sr.No	Name of Student	Enrollment No
1	Atharva Bakwad	24211000033

has successfully completed Industrial Training (315004) in **MITU Skillologies** from 25/05/2026 to 15/08/2026 for partial fulfilment towards completion of Diploma in Computer Engineering from Pimpri Chinchwad Polytechnic Institute code 0056.

Head of the Department

Mrs. M. S. Malkar

Head of the Institute

Dr. V. S. Byakod

Name of External Examiner

Seal of institute

ABSTRACT

Industrial training is one of the requirements to impart technical education at diploma level in computer engineering as per MSBTE. As such, the students are required to write and submit reports at the end of their training.

This report includes the engineering experiences, professional based practices and both TECHNICAL and NON-TECHNICAL knowledge and skilled development gained by me during industrial training at MITU Skillologies , Pune for 12 weeks from 25 May 2026 to 22 August 2026.

This industrial training consists of 10 chapters which altogether explain the topics, points to be covered during training. Chapter 1 and 2 explains the background study of industry, its organisational structure, responsibilities of different posts, its location etc. Chapter 3 explains about the major equipment's/machines in industry with their function specification and cost along with their maintenance. Chapter 4 consist of material handling and safety procedures. Chapter 5 consists about practical activities undertaken at workshop, common faults and their remedies with controlling procedure. Chapter 6 explains about special experiences encountered during training, recommendations and conclusion. And chapter 7 consists of the references

ACKNOWLEDGEMENT

We would like to express my gratitude to one and all who contributed me to success in the industrial training programme who helped me in various ways technical, management and all sort of guidance given to improve our professional and technical knowledge.

First, We would like to thank **Mr. Tushar Khute** of 'MITU Skillologies ' who allowed us for training experience.

We must give my special thanks to the Head of Department of computer engineering, respected **Mrs. M. S. Malkar** madam for her invaluable advice and constant encouragement to complete this seminar report in a successful manner then, **S.G.Aparanji** lecturer, who did all the co-ordination between students and company and thanks to other academic staff members for their great efforts for monitoring our training experience throughout the training period in order to ensure that we gain maximum knowledge out of it.

We would like to express my deep sense of gratitude to our esteemed Principal **Dr. V. S. Byakod** for her encouragement.

Our special thanks to all the supervisors and developers of 'MITU Skillologies ' who helped me by spending their valuable time and teaching me to resolved problems which we encountered while we was developing. These technical advices not only helped in resolving my problems but also improved our technical experience.

At last but not the least we would like to thank everyone who has made our training experience a pleasant and enjoyable one and helping to overcome the challenges we faced during training.

INPLANT TRAINING REPORT – MITU Skillologies

Sr.No	Name of Student	Enrollment No
1	Atharva Bakwad	24211000033

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Chapter 1: Organizational structure of Industry and General Layout

1.1 General Information Regarding the Organization:

Organization Name: Mitu Skillologies

Industry Type: IT Training and Skill Development Institute

Domain: Data Science, Artificial Intelligence, Machine Learning, Software Development, Python Programming, Web Development, Cloud Computing, and Emerging Technologies.

Location: 208, Adi Prime ,Moshi-Dehu Road ,Moshi Borhadewadi Pune, Maharashtra, India

Training Duration: 25/05/2026 to 22/08/2026

Training Program: Data Science, Artificial Intelligence & Machine Learning

MITU Skillologies is a leading Information Technology research, software development, and technical consulting organization located in Pune, Maharashtra. The organization specializes in Artificial Intelligence (AI), Machine Learning (ML), Data Science, Internet of Things (IoT), Cloud Computing, Software Development, and Open-Source Technologies. It provides industry-oriented training, software solutions, research support, internships, workshops, and technical consulting services to students, educational institutions, and industries. The organization focuses on bridging the gap between academic learning and industry requirements by providing practical knowledge through live projects, industrial training, and research-based learning. Students gain hands-on experience with modern technologies such as Python, Generative AI, Large Language Models (LLMs), Deep Learning, Computer Vision, Natural Language Processing (NLP), and cloud-based applications.

MITU Skillologies also promotes innovation, entrepreneurship, and open-source software development by encouraging students to work on real-world projects. With experienced trainers, researchers, and software professionals, the organization helps learners develop technical skills, problem-solving abilities, teamwork, and professional ethics required in today's IT industry.

1.2 Vision

The vision of **Mitu Skillologies** is:

- To become a globally recognized leader in Artificial Intelligence, Machine Learning, Data Science, Software Engineering, and emerging technologies.
- To provide high-quality, industry-oriented technical education, research, and innovative software solutions.
- To empower students, professionals, and organizations with advanced technical knowledge and practical skills.
- To promote innovation, creativity, research, and entrepreneurship through modern technology.
- To bridge the gap between academic learning and industrial requirements by delivering practical and project-based education.
- To create a strong ecosystem that supports technological advancement, digital transformation, and sustainable development.
- To contribute to society by developing skilled, ethical, and responsible technology professionals.

- To establish excellence in software development, technical consulting, and research at national and international levels.

1.3 Mission

The mission of **Mitu Skillologies** is:

- To provide high-quality technical education through experienced trainers and industry experts.
- To offer practical exposure using live industrial projects and real-world case studies.
- To develop analytical thinking, programming skills, and problem-solving capabilities among students.
- To encourage continuous learning and innovation in emerging technologies.
- To enhance students employability through career guidance, internships, placement assistance, and professional development programs.

1.4 Objectives

The major objectives of Mitu Skillologies are:

- To provide practical knowledge in Data Science and Artificial Intelligence.
 - To train students in Python programming and Machine Learning techniques.
 - To enhance technical and professional competencies required by the IT industry.
 - To develop analytical and logical thinking abilities.
 - To provide industrial exposure through real-time projects and internships.
-

- To improve communication, teamwork, and presentation skills.
- To prepare students for placements, higher education, and research opportunities.
- To encourage innovation and continuous skill development.

1.5 Organizational Structure

The organizational hierarchy of Mitu Skillologies ensures smooth management, effective training delivery, and proper coordination among departments.

Each department plays an important role in ensuring quality education, effective project execution, and successful completion of internship programs.

Chapter 2 : Introduction Of Industry/Organization

2.1 History OF Mitu Skillologies

MITU Skillologies is a technology-driven research, software development, and training organization established to provide quality technical education and industry-oriented solutions. The organization began its journey in Nashik and later established its headquarters in Pune, Maharashtra, with the objective of bridging the gap between academic learning and industrial requirements.

Since its establishment, MITU Skillologies has expanded its expertise in Artificial Intelligence, Machine Learning, Data Science, Software Development, Internet of Things (IoT), Cloud Computing, and Open-Source Technologies. It offers industrial training, internships, research guidance, software development, technical workshops, and consulting services to students, educational institutions, startups, and industries.

Over the years, the organization has collaborated with numerous engineering colleges and research institutions to promote innovation and practical learning. Through live projects, internships, and research activities, MITU Skillologies has helped thousands of students enhance their technical knowledge, programming skills, and professional competencies.

Today, MITU Skillologies is recognized as one of the leading technology training and research organizations in Maharashtra. With experienced trainers, software engineers, and researchers, it continues to contribute to skill development, innovation, and the advancement of modern technologies while preparing students for successful careers in the IT industry.

2.2 Type Of Products and Services Offered

MITU Skillologies provides a wide range of technology-driven products and services that support education, research, software development, and industrial innovation. The organization focuses on delivering practical, industry-oriented solutions using modern technologies and open-source platforms.

The major products and services offered by MITU Skillologies include:

- Artificial Intelligence (AI) and Machine Learning (ML) Solutions
- Data Science and Big Data Analytics Services
- Python Programming and Software Development
- Deep Learning and Computer Vision Applications
- Natural Language Processing (NLP) Solutions
- Generative AI and Large Language Model (LLM) Development
- Agentic AI and Intelligent Automation Systems
- Internet of Things (IoT) Solutions using Raspberry Pi and Embedded Systems
- Web Application and Mobile Application Development
- Cloud Computing and Database Management Solutions
- Industrial Training and Internship Programs
- Faculty Development Programs (FDPs)
- Corporate Training and Technical Workshops
- Research & Development (R&D) and Technical Consulting
- Project Development and Placement Assistance

MITU Skillologies follows a practical, project-based approach to learning and development. By integrating research, software engineering, and industry collaboration, the organization enables students and professionals to gain hands-on experience, improve technical skills, and develop innovative solutions for real-world challenges. It continuously updates its

services in line with emerging technologies to meet the evolving needs of industries and educational institutions.

2.3 STUDY OF SALES, TURNOVER, PROFITABILITY AND NUMBER OF EMPLOYEES

MITU Skillologies is a technology-driven organization specializing in research, software development, and industry-oriented training in Artificial Intelligence, Data Science, Machine Learning, and other emerging technologies. The organization is committed to providing practical education, technical consulting, and innovative software solutions for students, professionals, and industries.

Since its establishment, MITU Skillologies has expanded its activities in technical training, live projects, research, and software development. The organization focuses on developing industry-ready skills through practical learning, internships, workshops, and project-based education.

The organizational summary is as follows:

Year of Establishment: 2014

Head Office: Moshi, Pune, Maharashtra, India

Organization Type: Private IT Research, Software Development, and Technical Training Organization

Core Technologies: Artificial Intelligence, Machine Learning, Data Science, IoT, Software Development, and Open-Source Technologies.

Major Services: Industrial Training, Software Development, Research & Development, Technical Consulting, and Technical Workshops.

Internship Program: Industry-Oriented Internship with Live Projects.

Technical Team: Experienced Software Engineers, Researchers, Project Mentors, and Technical Trainers.

The organization does not publicly disclose its annual sales, turnover, profitability, or exact employee strength. However, through continuous research, software development, industrial training, and technology consulting, MITU Skillologies has established itself as a recognized organization in the fields of Artificial Intelligence, Data Science, and emerging technologies.

2.4 Areas of Application

The technologies taught at Mitu Skillologies are widely used in various industries.

- Banking and Finance
- Healthcare
- Retail and E-commerce
- Manufacturing
- Telecommunications
- Education
- Marketing Analytics
- Insurance
- Logistics

2.5 Technologies Used

During the internship, students were introduced to several industry-standard technologies.

Programming Languages

- Python
- SQL

Data Science Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn

Machine Learning Libraries

- Scikit-learn

Development Tools

- Jupyter Notebook
- Visual Studio Code
- Google Colab
- GitHub
- Streamlit

Database

- MySQL

2.6 Benefits of the Training

MITU Skillologies has established itself as a leading training organization in Artificial Intelligence and Emerging Technologies by focusing on practical learning, industry collaboration, and skill development. The organization has achieved several milestones in technical education and professional training.

Some of its major achievements are:

- More than 10 years of experience in providing quality technical education and professional training.

- Collaboration with 50+ MoU partner engineering colleges to strengthen industry-oriented learning.
- Successfully empowered 5,000+ alumni through training programs, internships, and certification courses.
- Adoption of a 100% practical hands-on learning approach with implementation-focused projects.
- Offering 6-month integrated internship programs that provide real-world project experience.
- Delivering training in 20+ open-source and emerging technologies, including Artificial Intelligence, Machine Learning, Data Science, Python, IoT, Deep Learning, and Generative AI.
- Conducting Faculty Development Programs (FDPs), workshops, seminars, and technical events for educational institutions.

2.7 Future Scope

Data Science and Artificial Intelligence are among the fastest-growing technologies in today's digital world.

Future opportunities include:

- AI Engineer
- Data Scientist
- Machine Learning Engineer
- Data Analyst
- Business Intelligence Analyst
- Deep Learning Engineer
- NLP Engineer
- Computer Vision Engineer
- Data Engineer
- AI Researcher

2.8 Summary

Mitu Skillologies plays an important role in preparing students for modern IT industries through industry-oriented training, live projects, and internship programs. The organization provides an excellent platform to learn emerging technologies such as Data Science, Artificial Intelligence, and Machine Learning. The knowledge and practical experience gained during the internship significantly enhanced technical expertise, problem-solving ability, and professional skills, making students industry-ready and well-prepared for future career opportunities.

Chapter 3: Types of Major Equipment/Instruments/ Machines Used in Industry

3.1 Hardware Infrastructure Used

3.1.1 Desktop Computers / Workstations

High-performance desktop computers were used during the training to perform programming, data analysis, machine learning model development, and project implementation. These systems were equipped with modern processors, sufficient RAM, and SSD storage to ensure smooth execution of computational tasks.

Applications:

- Python Programming
- Data Analysis
- Machine Learning Model Training
- Streamlit Deployment
- Database Management

3.1.2 Laptop Systems

Students also used personal laptops for coding, documentation, project development, and presentations.

Specifications (Typical):

- Intel Core i5/i7 Processor
- 8 GB or 16 GB RAM
- SSD Storage
- Windows 11 / Ubuntu Linux
- Internet Connectivity

3.1.3 High-Speed Internet and Wi-Fi Network

A reliable internet connection was essential for downloading datasets, installing Python libraries, accessing cloud-based notebooks, attending online sessions, and collaborating through GitHub.

Uses:

- Research Work
- Cloud Computing
- GitHub Repository Access
- Software Installation
- Online Learning Resources

3.1.4 Projector and Smart Classroom

Projectors and digital presentation systems were used by trainers to demonstrate coding techniques, machine learning concepts, project workflows, and technical presentations.

3.2 Software Used During Training

3.2.1 Python

Python was the primary programming language used during the internship because of its simple syntax, readability, and extensive support for Data Science and Artificial Intelligence. Python provides a large collection of built-in and third-party libraries that make programming, data processing, analysis, visualization, automation, and machine learning easier. During the training, Python was used to develop programs, perform data analysis, process datasets, create visualizations, and understand basic machine learning concepts. Libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn can be used with Python for various data-related tasks. Python was preferred because programs can be developed with fewer lines of code and its large community provides extensive learning resources and support. Its cross-platform compatibility also allows Python programs to run on different operating systems.

Features:

- Easy to learn
- Interpreted language
- Object-oriented
- Portable
- Open-source
- Dynamically typed
- Large standard library
- High-level language
- Extensible and embeddable
- GUI support

Applications:

- Programming
- Data Analysis
- Machine Learning
- Automation
- Data Visualization

Advantages:

- Easy syntax
- Large community support
- Rich library ecosystem
- Cross-platform compatibility

3.2.2 Jupyter Notebook

Jupyter Notebook was used as an interactive environment for writing, executing, and documenting Python programs. It allows code to be divided into separate cells and executed step-by-step, making it easier to understand the output of individual statements or sections of a program. Jupyter Notebook was particularly useful during data analysis because code, output, tables, graphs, and explanations can be maintained in the same notebook. It was used for data cleaning, Exploratory Data Analysis (EDA), data visualization, machine learning experiments, and project documentation. One of its major advantages is that changes can be tested immediately without executing the complete program. Markdown cells can also be used to add headings, explanations, notes, and documentation, making the notebook easy to understand and maintain.

Features:

- Interactive coding environment
- Step-by-step code execution
- Markdown documentation
- Visualization support
- Easy debugging

Uses:

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Machine Learning Experiments
- Project Documentation

3.2.3 Visual Studio Code (VS Code)

Visual Studio Code was used as the main Integrated Development Environment (IDE) for developing Python programs and the Streamlit application. It provides a complete environment where source code can be written, edited, executed, tested, and debugged. VS Code was useful for developing larger projects because files, folders, Python programs, libraries, and project resources can be managed in a single workspace.

Features such as IntelliSense/code completion, debugging tools, terminal support, Git integration, and extensions helped improve the development process. Python extensions can also provide syntax highlighting, error detection, code suggestions, and program execution. VS Code was therefore more suitable than a notebook environment for organizing and developing the complete project.

Features:

- Intelligent Code Completion
- Debugging Support
- Git Integration
- Extension Marketplace
- Terminal Support

3.2.4 Google Colab

Google Colaboratory (Google Colab) was used as a cloud-based environment for writing and executing Python programs and Jupyter notebooks. It works through a web browser, so Python does not need to be installed locally on the computer. Google Colab was useful for performing experiments, testing Python programs, working with datasets, and conducting machine learning activities. It also provides access to computational resources such as GPU support, which can be useful for computationally intensive machine learning tasks. Colab supports integration with Google Drive, making it convenient to store and access notebooks and datasets. It also makes collaboration easier because notebooks can be shared with other users, allowing them to view, execute, and modify the work when appropriate.

Advantages:

- Free GPU Support
- Cloud Storage Integration
- Easy Collaboration
- No Installation Required

3.3 Data Science Libraries

3.3.1 NumPy

NumPy (Numerical Python) is a Python library mainly used for numerical and mathematical operations. Its main feature is the NumPy array, which allows large amounts of numerical data to be stored and processed efficiently.

NumPy provides functions for mathematical calculations, statistical operations, matrix operations, linear algebra, and numerical analysis. It is also used as a foundation for several other Data Science and Machine Learning libraries.

Applications:

- Matrix Operations
- Statistical Calculations
- Linear Algebra
- Numerical Analysis

3.3.2 Pandas

Pandas is a Python library used for data manipulation, data preprocessing, and data analysis. It provides two important data structures: Series and DataFrame.

A DataFrame looks similar to a table containing rows and columns, which makes Pandas very useful for working with datasets. It can read data from files such as CSV and Excel, handle missing values, filter data, sort records, group data, and perform data transformations.

Functions Performed:

- Reading CSV Files
- Handling Missing Values
- Filtering Data
- Data Aggregation
- Data Transformation

3.3.3 Matplotlib

Matplotlib is a Python library used for data visualization. It converts numerical or analytical data into graphical representations, making patterns and trends easier to understand. It can create different types of charts and graphs depending on the type of data being analyzed.

Charts Created:

- Line Charts
- Bar Charts
- Histograms
- Scatter Plots
- Pie Charts

3.3.4 Seaborn

Seaborn is a Python library used for statistical data visualization. It is built on top of Matplotlib and provides a simpler way to create attractive and informative statistical graphs. Seaborn is particularly useful when we want to understand the relationship, distribution, correlation, and comparison between variables in a dataset.

Visualizations:

- Heatmaps
- Pairplots
- Boxplots
- Correlation Matrix
- Distribution Plots

3.4 Machine Learning Library

Scikit-learn

Scikit-learn is a popular Python library used for Machine Learning. It provides ready-made tools and algorithms for preparing data, training machine-learning models, making predictions, and evaluating model performance.

In your project, Scikit-learn can be used to build a Customer Lifetime Value (CLV) prediction model. It helps convert prepared customer data into a trained model that can predict the expected lifetime value of customers.

Applications:

- Data Splitting
- Feature Scaling
- Model Training
- Prediction
- Model Evaluation

Algorithms Studied:

- Linear Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Logistic Regression

TensorFlow

TensorFlow is an open-source Machine Learning and Deep Learning library developed by Google. It is used to build, train, and deploy machine-learning and neural-network models. **Applications:**

- Deep Learning
- Neural Networks
- Image Recognition

- Natural Language Processing
- Prediction

3.5 Database Management

Software MySQL

MySQL is a relational database management system (RDBMS) used to store, retrieve, and manage structured data.

Data in MySQL is organized into databases, tables, rows, and columns. SQL commands are used to perform operations on the stored data.

Operations Performed:

- Database Creation
- Table Creation
- Insert Records
- Update Records
- Delete Records
- SQL Queries

3.6 Version Control System

GitHub

GitHub is a platform used for hosting and managing software projects using Git version control. It allows developers to store source code online, track changes, maintain different versions, and collaborate with other developers.

In a project, GitHub can be used as a central repository for storing Python files, datasets or project documentation, depending on the repository requirements.

Features:

- Code Repository
- Version Tracking
- Team Collaboration
- Backup of Source Code

3.7 Deployment

Tool Streamlit

Streamlit is an open-source Python framework used to create and deploy interactive web applications for Data Science and Machine Learning projects. It allows Python developers to convert data-analysis programs and trained machine-learning models into user-friendly web applications without requiring extensive knowledge of HTML, CSS, or JavaScript.

In this training project, Streamlit was used to deploy the Customer Lifetime Value (CLV) Prediction model as a web application. The application allows the user to enter customer-related information, submit the data, and obtain the predicted CLV through an interactive interface.

Features:

- Interactive User Interface
- Real-Time Predictions
- Easy Deployment
- User-Friendly Dashboard

Applications:

- Customer Data Input
- CLV Prediction

Chapter 4: Safety Procedures Followed and Safety Gear Used

4.1 Introduction

Safety is one of the most important aspects of every organization, including IT training institutes and software development companies. A safe working environment protects employees, students, equipment, and data from accidents, hazards, and security threats. During the industrial training at **Mitu Skillologies**, various safety measures and workplace guidelines were followed to ensure a secure, healthy, and productive learning environment.

Since the training mainly involved computer-based activities such as programming, data analysis, machine learning model development, and software implementation, greater emphasis was placed on **computer laboratory safety, electrical safety, cyber security, data protection, and ergonomic practices**.

4.2 Importance of Safety in IT Industry

Following safety procedures provides several benefits:

- Prevents accidents and injuries.
- Protects computer systems and electronic equipment.
- Ensures data privacy and information security.
- Reduces hardware failures.
- Maintains a healthy working environment.
- Improves productivity and work efficiency.
- Promotes professional discipline and responsibility.

4.3 Safety Procedures Followed During Training

The following safety procedures were followed throughout the internship:

1. Electrical Safety

- Electrical equipment was inspected before use.
- Loose electrical connections were avoided.
- Wet hands were never used while handling electrical devices.
- Computers were properly switched off after use.
- Power sockets were not overloaded.
- UPS systems were used to protect computers from sudden power failures.

2. Computer Laboratory Safety

- Students maintained a clean and organized workspace.
- Food and drinks were not allowed near computer systems.
- Computers were handled carefully to prevent physical damage.
- Cables and networking devices were arranged properly to avoid accidents.
- Computer systems were shut down correctly after every practical session.

3. Internet and Cyber Security

As the training involved online platforms and cloud-based tools, cyber security practices were strictly followed.

These included:

- Using strong passwords.
- Regular password updates.
- Installing antivirus software.
- Avoiding suspicious websites and downloads.
- Protecting confidential project data.
- Secure login to online platforms.
- Logging out after completing work.

4. Data Security

The project datasets and source code were protected by following standard data management practices.

These included:

- Maintaining backup copies of project files.
- Using GitHub for version control.
- Avoiding unauthorized access to project folders.
- Organizing files systematically.
- Preventing accidental deletion of important data.

5. Fire Safety

Basic fire safety measures were available within the institute.

These included:

- Fire extinguishers installed at accessible locations.
- Emergency exits clearly marked.
- Immediate reporting of electrical faults.
- Safe evacuation procedures during emergencies.

4.4 Safety Gear Used

Although Mitu Skillologies is an IT training organization and does not require industrial protective equipment such as helmets or gloves, the following safety facilities were available:

- Fire Extinguishers
- First Aid Box

- UPS (Uninterruptible Power Supply)
- CCTV Surveillance System
- Smoke Detectors
- Emergency Exit Signs
- Proper Electrical Earthing
- Air Conditioning and Ventilation System

These facilities ensured a safe and comfortable learning environment.

Chapter 5: Particulars of Practical Experiences in Industry

5.1 Introduction

The Industrial Training at Mitu Skillologies provided practical exposure to Data Science, Artificial Intelligence (AI), and Machine Learning (ML). During the 12-week training program, various theoretical concepts learned in academics were implemented through practical assignments, coding exercises, case studies, and a real-time industrial project. The training followed a hands-on approach, enabling students to understand the complete workflow of a Data Science project, from data collection to model deployment. Throughout the internship, emphasis was placed on developing programming skills, analytical thinking, data visualization techniques, machine learning model development, and professional software development practices. The training also helped in understanding industrial work culture, teamwork, documentation, and project management.

Topics Covered During Training

The following topics were covered during the industrial training:

5.2 Python Programming

5.3 Data Analysis using NumPy and Pandas

5.4 Exploratory Data Analysis (EDA)

5.5 Machine Learning

5.2 Python Programming

The training began with Python programming, which is one of the most popular programming languages used in Data Science and Artificial Intelligence.



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The following concepts were practiced:

- Variables and Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- Lists, Tuples, Sets, and Dictionaries
- File Handling
- Exception Handling
- Object-Oriented Programming (OOP)
- Modules and Packages

Practical programs were developed to strengthen logical thinking and programming skills.

5.3 Data Analysis using NumPy and Pandas

After learning Python, practical sessions focused on data analysis using NumPy and Pandas.

Activities Performed

- Importing datasets
- Reading CSV and Excel files
- Data cleaning
- Handling missing values
- Removing duplicate records
- Data filtering
- Sorting
- Data aggregation
- Grouping
- Feature creation



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- Statistical analysis

These activities helped in understanding how raw industrial data is converted into meaningful information.

5.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of datasets before applying Machine Learning models.

The following visualizations were created:

- Histogram
- Bar Chart
- Scatter Plot
- Box Plot
- Heatmap
- Pair Plot
- Correlation Matrix

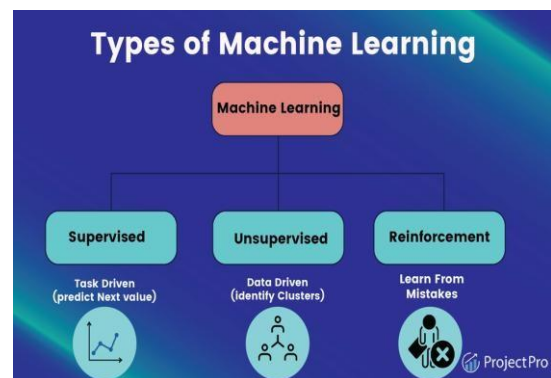
EDA helped identify patterns, trends, outliers, and relationships among different variables.

5.5 Machine Learning

The internship introduced the fundamentals of Machine Learning and its industrial applications.

Topics Covered

- Introduction to Machine Learning
- Supervised Learning
- Unsupervised Learning
- Regression
- Clustering



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- Model Training
- Model Testing
- Cross Validation
- Hyperparameter Tuning

Students learned how machine learning models are developed and evaluated using real-world datasets.

5.6 Major Project: Employee Attrition Prediction

The major practical activity carried out during the internship was the development of a Machine Learning project titled "Employee Attrition Prediction." The project aims to predict whether an employee is likely to leave the organization based on Human Resource (HR) data. This helps the HR department identify employees at risk of attrition and take preventive actions to improve employee retention and reduce turnover.

5.7 Project Workflow

The project was developed using the following stages:

Step 1: Data Collection

The dataset was collected from the IBM HR Analytics Attrition Dataset available on Kaggle. It contains employee information such as age, department, job role, income, work experience, overtime, job satisfaction, work-life balance, and attrition status.

Step 2: Exploratory Data Analysis (EDA)

The collected dataset was analyzed before model development. The following tasks were performed:

- Checked for missing values.
- Removed unnecessary constant columns.
- Analyzed employee attrition distribution.
- Compared attrition based on Department, Job Role, and Overtime.
- Created box plots for Age, Monthly Income, Years at Company, and Distance from Home.
- Generated a correlation heatmap to study relationships among features.

Step 3: Data Preprocessing

The dataset was prepared for machine learning by performing:

- Removing unnecessary columns.
- Encoding categorical variables.
- Standardizing numerical features.
- Handling class imbalance using SMOTE or `class_weight='balanced'`.
- Splitting the dataset into 80% training and 20% testing sets.

Step 4: Machine Learning Model Development

Different machine learning algorithms were implemented and compared, including:

Logistic Regression

Decision Tree

Random Forest

The models were trained using historical employee data to predict employee attrition.

Step 5: The model performance was evaluated using:

Accuracy

Recall

F1-Score

ROC-AUC

Confusion Matrix

These evaluation metrics were used to identify the best-performing model for employee attrition prediction.

Step 6: Streamlit Deployment

The final model was deployed using Streamlit. The application allows HR personnel to enter employee details and receive:

Employee Attrition Risk (Low/Medium/High)

Top 5 Risk Factors influencing attrition

This deployment demonstrates the practical use of Machine Learning in Human Resource management.

5.8 Overall 12-Week Internship Experience

During my 12-week industrial training at MITU Skillologies, Pune, I gained practical exposure to Data Science, Artificial Intelligence, Machine Learning, Python programming, data analysis, and project development. The training gave me an opportunity to understand how theoretical concepts are applied in practical situations.

Throughout the internship, I participated in practical sessions, coding activities, data-analysis exercises, Machine Learning experiments, and project-related activities. I worked with tools and technologies such as Python, Jupyter Notebook, Google Colab, VS Code, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, MySQL, GitHub, and Streamlit.

During the initial phase of the training, I focused on developing my Python programming skills and understanding programming concepts through practical exercises. Later, I worked with datasets using NumPy and Pandas and learned how to clean, transform,

analyze, and visualize data. I also performed Exploratory Data Analysis to identify patterns, relationships, trends, and outliers in datasets.

As the training progressed, I learned the fundamentals of Machine Learning and studied algorithms such as Linear Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Logistic Regression. I gained experience in preparing datasets, splitting data into training and testing sets, training models, making predictions, and evaluating model performance.

One of the important parts of my internship was working on the Customer Lifetime Value (CLV) Prediction project. This helped me understand the complete workflow of a Data Science and Machine Learning project, starting from data preparation and analysis and continuing through model development, prediction, and deployment. I used Streamlit to create an interactive web application for the project.

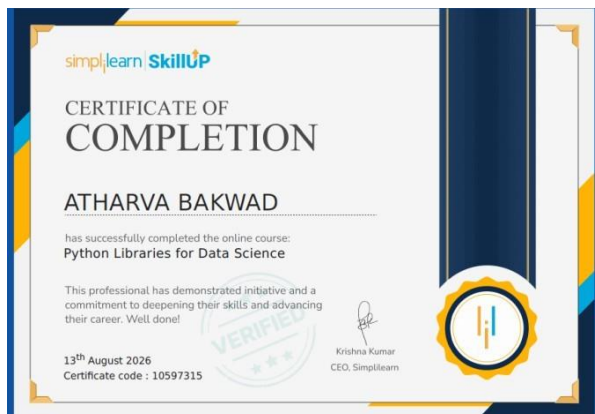
The internship also helped me improve my problem-solving, debugging, documentation, teamwork, communication, and time-management skills. I learned how to identify errors, search for suitable solutions, organize project files, use version control, and present technical work.

Overall, my 12-week internship was a valuable practical learning experience. It helped me connect the concepts learned during my diploma studies with real-world Data Science and Machine Learning applications. The training increased my confidence in Python programming, data analysis, Machine Learning, and project development and gave me a better understanding of how a technical project is developed from raw data to a working application.

Chapter 6: Details for Online Workshop

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Sr. No	Tittle	Duration	Organization/Industry
1	Introduction of Data science	1 week	Simplilearn
2	Machine learning	40 hrs	Simplilearn
3	Data Visualization	3 days	Simplilearn
4	Scikit-Learn	4 days	Simplilearn
5	Python Libraries For Data Science	3 weeks	Simplilearn
6	Python Programming	1 month	Mitu Skillologies
7	Machine Learning Using Python	60 hrs	Simplilearn



Chapter 7: References

7.1 REFERENCES

The following books, websites, software documentation, and online learning resources were referred to during the industrial training for learning Artificial Intelligence, Machine Learning, Data Science, Python Programming, and completing the Employee Attrition Prediction project.

MITU Skillologies Official Website

<https://mitu.co.in/>

Python Official Documentation

<https://docs.python.org/3/>

NumPy Documentation

<https://numpy.org/doc/>

Pandas Documentation

<https://pandas.pydata.org/docs/>

Scikit-learn Documentation

<https://scikit-learn.org/stable/documentation.html>

Matplotlib Documentatio

<https://matplotlib.org/stable/>

Jupyter Notebook Documentation

<https://jupyter.org/documentation>

Visual Studio Code Documentation

<https://code.visualstudio.com/docs>

Streamlit Documentation

<https://docs.streamlit.io/>

GitHub Documentation

<https://docs.github.com/>

IBM HR Analytics Attrition Dataset (Kaggle)

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

OpenAI – ChatGPT

<https://chat.openai.com/>

Student Github Link :-

Atharva Bakwad